FACE AND IDENTITY VERIFICATION USING CLOUD BASED SAAS MODELS

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CSE4297 Engineering Project 1

Project Specification Document

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1. PROBLEM STATEMENT

Authentication processes have become an integral part of our daily lives with the proliferation of e-government systems, online banking, healthcare portals, public transportation systems and smart buildings. However, widely used methods such as passwords, access cards, or single-factor biometric systems harbor various security risks and usability issues. Passwords can be stolen, cards can be cloned, and simple biometric systems can be easily circumvented by spoofing attacks using static photos or videos. This situation creates significant security gaps for both individual users and corporate entities, establishing the clear importance and worth of developing more robust solutions.

2. PROBLEM DESCRIPTION AND MOTIVATION

Contemporary biometric systems often lack the advanced liveness detection mechanisms required to counter increasingly sophisticated spoofing attempts. The absence of multi-modal biometric fusion further exposes these systems to presentation attacks, while their reliance on centralized data storage architectures renders them attractive targets for cybercriminals. Moreover, integration challenges between physical access control infrastructures and digital authentication platforms contribute to fragmented and heterogeneous security ecosystems that are difficult to monitor, manage, and scale effectively.

Many legacy authentication systems fundamentally violate essential software engineering principles, most notably the Single Responsibility Principle (SRP) within the SOLID framework which has five fundamental object-oriented design guidelines that help create software systems that are flexible, maintainable, and easy to extend. This principle dictates that each software module must possess exactly one well-defined responsibility. However, the violation of this standard frequently results in monolithic services that encapsulate disparate functionalities, ranging from user management to audit logging. This architectural flaw creates rigid, complex, and unmaintainable codebases.

Beyond these architectural constraints, conventional authentication services also encounter significant performance limitations in distributed environments. This is especially critical when handling a high volume of simultaneous verification requests. The absence of effective caching mechanisms, poor load balancing strategies, and

inefficient database queries contribute to these bottlenecks. Together, these factors lead to increased latency and result in a negative user experience.

However, biometric security now faces a major new challenge from Generative AI, going beyond traditional tricks like printed photos or masks. Attackers today use advanced technology to create "Deepfakes" that look incredibly real. These computer-generated images and videos can copy tiny facial movements and voice tones with dangerous accuracy. Standard security systems were originally designed to spot static fake items. However, they often struggle to tell the difference between these active, AI-generated fakes and real human users. This creates a constant race between security teams and attackers.

Therefore, the primary motivation for this project is to create a secure, flexible, and user-friendly platform that consolidates these fragmented and insecure authentication methods. We plan to develop a cloud-native Software as a Service (SaaS) platform targeting Business-to-Business (B2B) and Business-to-Business-to-Consumer (B2B2C) markets. The project aims to unify both physical and digital access control for a wide range of applications, from residential complexes to corporate offices. The platform is intended to differentiate itself and fill a significant market gap through its unique liveness detection mechanism against fraud and its high accuracy achieved by fusing multiple biometric modalities.

3. MAIN GOAL AND OBJECTIVES

The project aims to develop a multi-platform, multi-tenant, and cloud-native SaaS platform for secure and flexible multi-biometric identity authentication.

The specific objectives set to achieve this main goal are as follows:

- ***Objective 1:*** To develop an innovative and active liveness detection algorithm. The “Biometric Puzzle” our original liveness verification concept requires users to complete a randomly generated sequence of actions (e.g., smiling, blinking) to prevent spoofing attacks using static photos, video recordings, or masks.
- ***Objective 2:*** To establish a scalable, multi-tenant, cloud-native Software as a Service (SaaS) platform.
  This goal involves designing a microservices architecture where the Identity Core API manages tenants and roles, while the Biometric Processor API

handles heavy biometric computations, ensuring the system remains fast, secure, and stable for multiple corporate clients simultaneously.

- ***Objective 3:*** To develop a user-friendly mobile application.
  A cross-platform mobile application for Android and iOS that enables end-users to enroll their biometric data and authenticate their identity.

- ***Objective 4:*** To verify identities using various document types.
  To enable the automatic extraction of facial data from embedded photographs in ID cards, passports, driver's licenses, and student/institutional cards.

4. RELATED WORK

Biometric authentication, particularly face recognition and liveness detection, has progressed rapidly with deep learning–based approaches. Foundational models such as **DeepFace** [1] and **FaceNet** [2] introduced embedding-based identity verification but relied on monolithic, device-bound architectures that limited scalability. To better understand how modern systems address these limitations, both academic and industrial face recognition and liveness detection solutions have been evaluated in detail.

4.1 Advances in Deep Face Recognition Models

Facial representation learning advanced significantly with approaches such as **ArcFace** [3], which introduced additive angular margin loss to improve class separability and set a new benchmark in verification accuracy. Subsequent models including **CosFace** [4], **SphereFace** [5], **MagFace** [6], and **AdaFace** [7] further enhanced robustness through margin-based and quality-aware strategies designed to handle low-quality, blurred and off-angle captures.

Although these methods achieve strong performance in unconstrained environments, most studies focus on algorithmic improvements rather than deployment challenges. In our platform which we named as **FIVUCSAS**, these state-of-the-art models are integrated via the **DeepFace** framework inside a containerized inference microservice supporting GPU acceleration and communicating with the Identity Core through REST APIs and asynchronous message queues. This architecture enables scalable and low-latency biometric processing while ensuring reliable multi-tenant operation, effectively

bridging the gap between research-level face recognition models and production-ready SaaS platforms.

4.2 Liveness Detection: Passive, Active, and Hybrid Approaches

Liveness detection has become essential in biometric verification, particularly with the rise of deepfake and replay attacks. Traditional passive methods rely on texture and illumination analysis using CNN models such as **ResNet-18**, **MobileNetV3**, and **Xception**, achieving strong performance on presentation attack datasets [8]. Active approaches complement these techniques through challenge–response actions like blinks, head movements, or facial gestures measured with Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) metrics, while frameworks like **MediaPipe** [9] Face Mesh enable real-time landmark tracking on mobile devices. However, many of these solutions function as isolated models without an integrated workflow. Our system addresses this gap through a hybrid two-stage liveness pipeline combining passive CNN-based spoof detection with MediaPipe-driven active verification in the Biometric Puzzle (blink, smile, head movement). This design provides strong resilience against photo, video, and deepfake attacks, maintains low-latency performance, and aligns with ISO/IEC 30107-3 Presentation Attack Detection (PAD) standards.

4.3 Identity Document Verification, NFC, and ICAO Standards

Beyond face recognition, many security-critical applications require matching a selfie with an identity document. Studies such as DocFace introduced specialized ID–selfie matching pipelines, while other approaches combine OCR, edge detection, and MRZ parsing for document authenticity.

In national identity systems, standards like **Near Field Communication (NFC)** passport chip reading and **International Civil Aviation Organization (ICAO)** specifications for **electronic Machine-Readable Travel Documents (eMRTDs)** ensure secure access to embedded biometric data. Commercial systems such as **BioGATE Pass** leverage these ICAO-compliant NFC workflows to provide high trust identity verification.

Our system supports official ID photo comparison and is designed to align with ICAO principles. Although full NFC passport workflows are planned, the modular architecture already enables easy integration of document verification services and additional biometric modalities.

4.4 Comparison of Industrial Solutions

To contextualize the capabilities of our system, we compare it with prominent commercial and open-source solutions. The following table summarizes key features:

Feature	Azure Face Live.	Amazon Rekognition	Sodec Technologies	FaceAuth Vision	BioGATE Pass	FIVUCSAS
Open Source	✗	✗	✗	✓	✗	✓
Liveness	✓	✓	✗	✗	✓	✓
Offline Support	✗	✗	✓	✓	✗	✓
Multi-Tenant	✓	✓	✗	✗	✗	✓
Multi-Platform	✓	✓	✓	✓	✗	✓
GPU Acceleration	✓	✓	✗	✗	✗	✓
PAD Compliance	✓	✓	✗	✗	✓	✓

Table 1: Comparison of Face Recognition and Liveness Systems

Azure Face Liveness [10] and **Amazon Rekognition** [11] offer highly accurate PAD-compliant facial verification but are fully proprietary, cloud-only, and lack offline capabilities. They are not easily customizable for educational or enterprise tenants requiring privacy-preserving or on-premises configurations.

Sodec Technologies [12] provides regional ID verification for e-government and banking but lacks multi-tenant support and advanced liveness strategies. **FaceAuth Vision,** being open-source and OpenCV-based, enables offline usage and edge deployments but lacks modern anti-spoofing methods and enterprise-grade scalability.

BioGATE Pass [13] integrates NFC passport chip reading under ICAO compliance, enabling high-assurance verification, yet it does not support multi-tenant SaaS deployments or multi-platform integration.

Compared to existing systems, **FIVUCSAS** uniquely combines cloud-native, multi-tenant identity management with multi-platform clients. Featuring GPU-accelerated microservices, hybrid liveness detection, and offline edge capabilities, its open-source flexibility positions the system as a bridge between research frameworks and commercial enterprise solutions.

4.5 Cloud-Native Biometric Verification and Multi-Tenant Architectures

Recent research underscores the necessity of cloud-native microservices for scaling biometric workloads, emphasizing modular decomposition and vector-based storage [14]. However, few existing implementations successfully combine biometric computation, identity management, and tenant isolation within a single ecosystem [15].

FIVUCSAS addresses this gap by adopting a Hexagonal Architecture that separates domain logic via the Identity Core API, supported by event-driven communication using Redis. The system leverages dedicated GPU/CPU inference microservices for heavy computation and pgvector for high-speed embedding retrieval, providing complete support across browser, mobile, and offline-edge desktop clients. This design ensures consistent performance under high load and robust tenant isolation, while maintaining full extensibility for future modalities such as fingerprint or iris recognition.

5. SCOPE

The scope of this project includes the development of a Minimum Viable Product (MVP) of the platform, encompassing its core functionalities. The items that are within and outside the scope of this project are clearly defined below.

5.1. In Scope

- ***Backend Development:*** Development of the Identity Core API and Biometric Processor API microservices using Spring Boot and FastAPI.
- ***Database Architecture:*** Design of a multi-tenant data model on PostgreSQL, including user, role, and tenant tables, and the storage of biometric vectors using the pgvector extension.
- ***Face Biometrics:*** Integration of face recognition (1:N) and face verification (1:1) functionalities using the DeepFace library.
- ***Liveness Detection:*** Development of the on-device "Biometric Puzzle" algorithm using Google MediaPipe.
- ***Mobile Application:*** Development of basic user interfaces for both Android and iOS platforms using Flutter, enabling end-user enrollment and authentication.

5.2. Out of Scope

- ***Hardware Production:*** Design and manufacturing of dedicated Edge devices for physical access control are excluded. These interactions will be simulated using a Raspberry Pi or Jetson Nano.

5.3. Constraints

- ***Data Constraints:*** The system's accuracy will be limited by the performance of the pre-trained models provided by the DeepFace library.
- ***Image Quality Constraint:*** For optimal performance in face recognition and liveness detection, input images should have a resolution higher than 480p. Lower-quality images may reduce detection accuracy and reliability.

5.4. Assumptions

- It is assumed that stable versions of the chosen open-source libraries (DeepFace, MediaPipe, Spring, FastAPI) will be accessible and compatible throughout the project lifecycle.
- It is assumed that the mobile devices used for testing meet the minimum hardware requirements for real-time video analysis.

6. METHODOLOGY AND TECHNICAL APPROACH

The system employs a microservices-based architecture that ensures modularity, scalability, and fault isolation while forming a cloud-native multi-biometric SaaS platform with maintainability, resilience, and secure tenant isolation.

The specific structural components and interactions realizing this architecture are detailed below.

6.1. High-Level System Architecture

The platform follows a distributed microservices model operating behind a unified API Gateway. Services communicate via REST APIs for synchronous requests and Redis-based message queues for asynchronous workflows. The data layer combines relational and vector databases for structured and biometric data management.

Figure 1 illustrates the comprehensive flow of our system, detailing the interactions between system components. It highlights the step-by-step process for liveness detection, face image enrollment, and embedding generation.

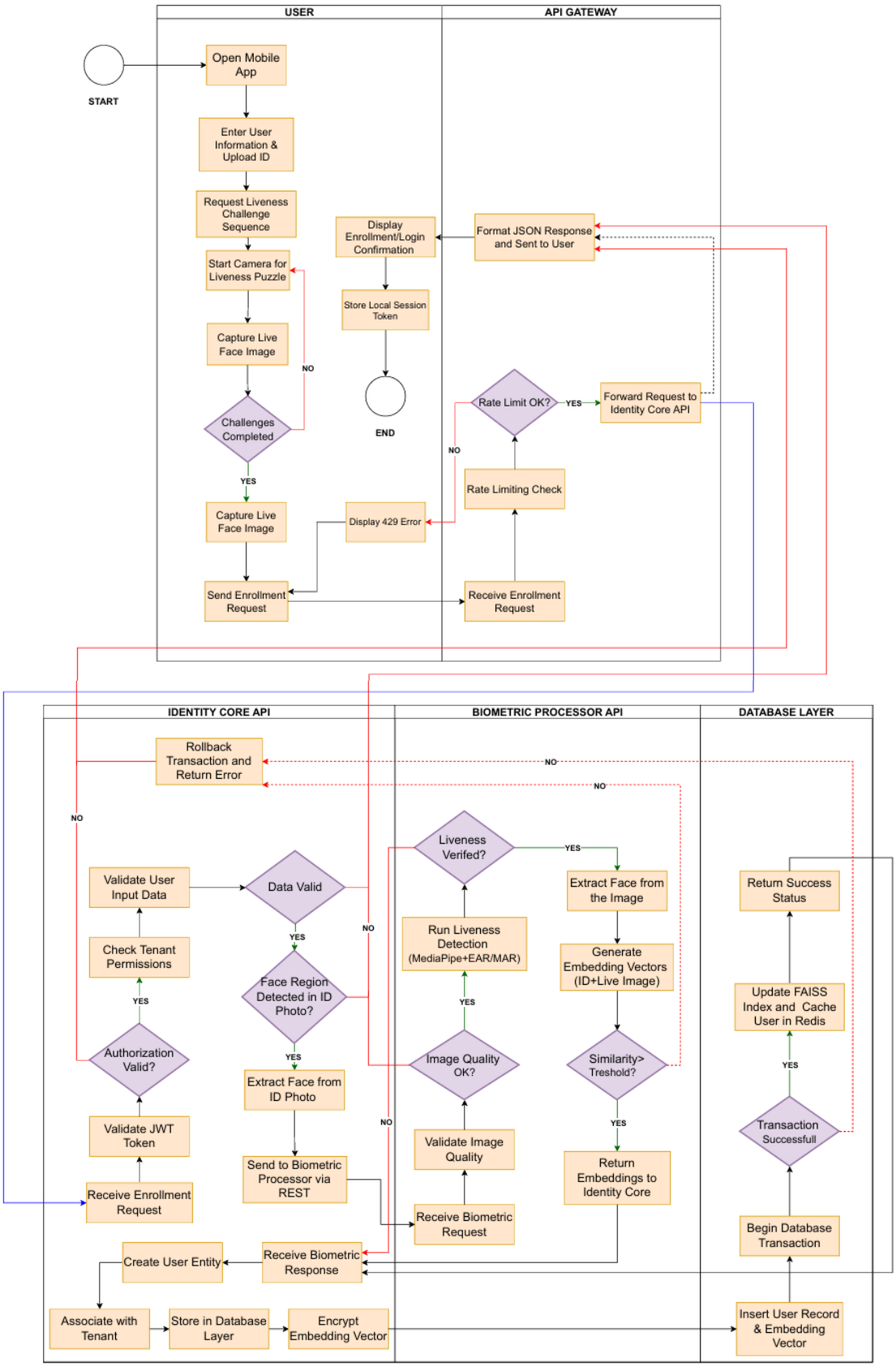

Figure 1: Flow Chart of Our System

The **Identity Core API**, developed with Spring Boot, follows a Hexagonal Architecture structure and applies Domain-Driven Design principles to maintain a clean separation between business logic and infrastructure. It is responsible for several core platform functions. These include managing **authentication and authorization** through JWT-based access tokens, refresh tokens, and role-based access control (RBAC). It also handles **tenant management**, ensuring strict multi-tenant data isolation and enabling tenant-specific configurations. In addition, it provides **user and role administration** by supporting full CRUD operations for users, roles, and permission assignments. Acting as the central coordination layer, the Identity Core API orchestrates communication between the frontend applications and the Biometric Processor API, managing the overall workflow across services.

The **Biometric Processor API**, implemented with FastAPI, is designed as an event-driven microservice optimized for compute-intensive machine learning operations. Built using Clean Architecture principles, it isolates biometric processing logic from infrastructure concerns while maintaining high performance. Its primary responsibilities include **feature extraction** through deep learning models to generate facial embeddings, **identity matching** for both 1:1 verification and 1:N identification using cosine similarity metrics, and performing multi-modal **liveness detection**. This liveness analysis combines passive and active cues to detect spoofing attempts and ensure that only genuine users pass the verification process.

The **Database Layer** relies on PostgreSQL as the primary datastore for transactional operations, providing full ACID compliance ensuring atomicity, consistency, isolation, and durability of transactions and enforcing strict multi-tenant isolation through Row-Level Security (RLS). The pgvector extension supports high-performance biometric search capabilities, enabling sub-millisecond similarity queries on 2,622-dimensional facial embeddings using HNSW indexing. This allows the system to efficiently handle both verification and identification scenarios.

The **Mobile Layer** is built using Flutter, implementing the Model-View-ViewModel (MVVM) architecture with the Business Logic Component (BLoC) pattern for predictable state management. The mobile app performs on-device liveness detection using Google MediaPipe, reducing inference latency from approximately 500 ms to 50 ms while enabling real-time facial landmark tracking. Biometric data is securely

transmitted through encrypted channels, and the Repository Pattern abstracts backend communication while preserving offline functionality for critical user flows.

The **Infrastructure Layer** uses a containerized microservices architecture with Docker Compose for development and Kubernetes-ready deployment manifests for production, ensuring easy scaling, portability, and consistent environments. It implements horizontal auto-scaling through Horizontal Pod Autoscaler (HPA), a service mesh for efficient inter-service communication, and centralized logging for observability and troubleshooting.

6.2. Technical Approach and Algorithms

Building on this general technical framework, the next subsection explains the first major component of the system: the liveness detection algorithm.

6.2.1. Biometric Puzzle (Liveness Detection) Algorithm

Liveness detection runs on the **mobile device** to improve responsiveness and privacy through an **active challenge–response** combined with **passive texture cues**:

1. The backend sends random challenges (e.g., blink, smile, turn head).
2. The app captures video via **OpenCV** and processes frames with **Google MediaPipe** to track 468 facial landmarks. Figure 2, illustrating facial geometry mapping.
3. Metrics like **Eye Aspect Ratio (EAR)** and **Mouth Aspect Ratio (MAR)** detect specific actions.
4. Once all challenges succeed, the app confirms liveness and securely transmits the validated frame to the backend.

This hybrid approach, illustrated in *Figure 3*, prevents spoofing attempts.

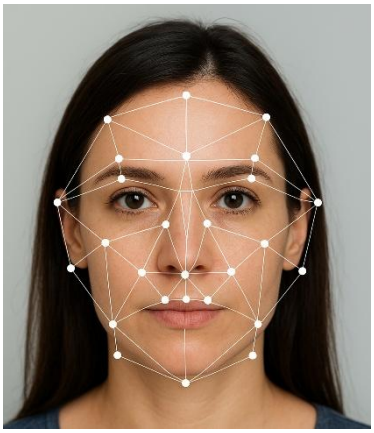

Figure 2: Face Landmarking

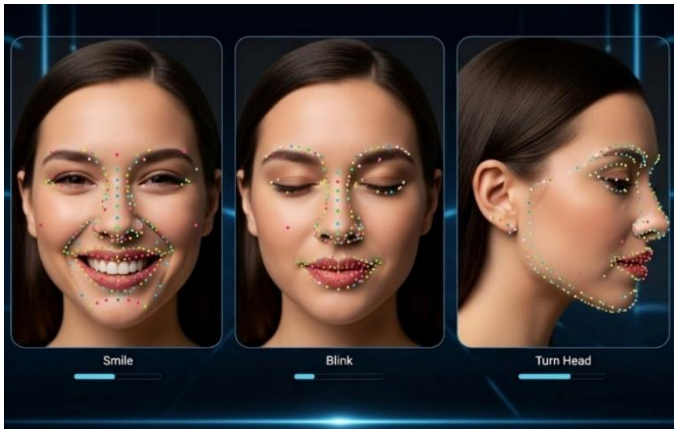

Figure 3: Biometric Puzzle, Facial Action Sequence

6.2.2. Face Recognition and Identity Verification

During enrollment, users may upload different types of identity documents (ID cards, passports, driver's licenses, student cards). The system extracts the face image and convertes into a 2,622-dimensional embedding via **DeepFace.represent()**, stored in a pgvector column. These extracted embeddings are paired with liveness-verified selfie images to enable reliable **ID-to-Selfie** matching during identity verification.

To quantify the similarity between two facial embeddings, the system uses cosine distance, a widely adopted metric in face recognition for measuring how close two feature vectors are in high-dimensional space. 0 indicates identical faces, while higher distance means less similarity. Equation 1 illustrates the cosine distance.

$$D_{\cos}(A, B) = 1 - \frac{A \cdot B}{\| A \| \| B \|} \qquad D_{\cos} \leq \text{threshold}$$

Equation 1: Cosine Distance and Verification Rule

The threshold (τ=0.68) was empirically optimized on the Labeled Faces in the Wild (LFW) validation set to achieve the optimal balance between False Acceptance Rate (FAR) and False Rejection Rate (FRR), minimizing the Equal Error Rate (EER). If the computed similarity is below or equal to this threshold, verification succeeds.

For **1:N** search scenarios, the platform uses **FAISS** for large-scale vector retrieval. Facebook AI Similarity Search (FAISS) is an open-source library designed for efficient similarity search and clustering of high-dimensional vectors.

Figure 4 illustrates the end-to-end face recognition pipeline.

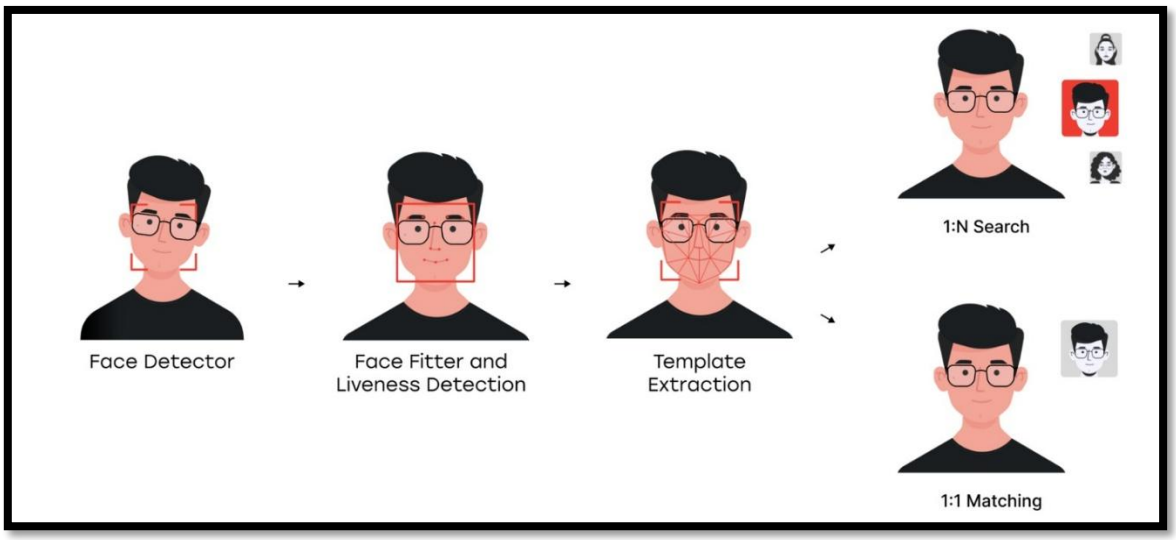

Figure 4: Recognition Pipeline Steps

6.2.3. System Integration and Security

All inter-service communication will occur over HTTPS with token-based authentication. Sensitive biometric embeddings are encrypted before storage, and role-based access control (RBAC) ensures that tenants can only access their isolated data domain.

Continuous integration and testing pipelines verify API integrity after each deployment using Postman and Pytest/JUnit test suites.

After outlining the core algorithms and processing techniques, the next section evaluates the system's performance to assess its accuracy, efficiency, and robustness under real-world conditions.

6.3. Performance Evaluation

System performance will be assessed for both biometric accuracy and operational efficiency:

- **Biometric Metrics:** False Acceptance Rate (*FAR*) < 1%, False Rejection Rate (*FRR*) < 3%, Equal Error Rate (*EER*) ≈ 3%
- **Liveness Metrics:** Evaluated using spoofing tests (printed photos, videos, masks) with Attack Presentation Classification Error Rate (*APCER)* and Bona Fide Presentation Classification Error Rate (*BPCER)* metrics.
- **System Metrics:** Average API response < 300 ms, Query latency < 100 ms, Stable throughput under simulated multi-tenant load.

6.4. Resources

- **Software & Development Tools:** IntelliJ IDEA (Java), PyCharm (Python), Visual Studio Code (Flutter), Docker Desktop, Postman, Git and GitHub for version control.
- **Libraries / Frameworks:** Spring Boot, FastAPI, Flutter, DeepFace, InsightFace, Google MediaPipe, OpenCV, PostgreSQL, Redis, FAISS, pgvector.
- **Hardware:** Personal computers for development. Two Android and two iOS devices for mobile testing and performance comparison.

Data: Publicly available datasets such as Labeled Faces in the Wild (LFW) and controlled test images collected with consent from project members. Synthetic data will be generated to simulate enrollment and authentication events.

7. PROFESSIONAL CONSIDERATIONS

7.1. Methodological considerations/engineering standards

The project follows modern software engineering principles to ensure quality, maintainability, and reproducibility. Project management will be conducted via a Kanban board on GitHub Projects, with source control and versioning managed through Git within a GitHub Organization. All code changes will undergo peer review through pull requests before merging. System design and documentation will use UML diagrams to visualize workflows, while a Gantt chart will track progress. Language-specific linting and formatting rules will ensure code quality, and testing frameworks will verify functionality and reliability.

7.2. Realistic Constraints

The project design has been done considering economic, environmental, ethical, health and safety, sustainability, and social factors.

- **Economic:** Open-source technologies eliminate software costs, and the SaaS model enables scalable, low-cost deployment. The system also reduces operational expenses by minimizing time, eliminating paper-based workflows, and removing the need for physical storage rooms for documents.
- **Environmental:** As a software system, it generates no direct waste. Energy efficiency is prioritized to reduce the indirect impact of server and device usage.
- **Ethical:** Users are informed and provide explicit consent for biometric data processing.
- **Health & Safety:** All personal and biometric data are encrypted using TLS and AES-256. The system poses no physical risks.
- **Sustainability:** The modular microservices architecture and Docker-based deployment support maintainability and long-term scalability.
- **Social:** The platform enhances security and user experience while enforcing strict data access controls to prevent misuse for surveillance.

7.3. Legal considerations

As the platform processes biometric data, it will comply with the Turkish Personal Data Protection Law (KVKK No. 6698) and the EU General Data Protection Regulation (GDPR). Data handling will adhere to principles of data minimization, purpose

limitation, explicit consent, security, and the right to be forgotten. All open-source dependencies will be checked for license compatibility to prevent legal conflicts in future commercialization. Documentation will include each dependency's license and compliance status.

8. MANAGEMENT PLAN

8.1. Description of Task Phases

8.1.1 Requirements Analysis & Literature Review

- **Academic Literature Survey** includes an examination of contemporary facial recognition architectures such as DeepFace, FaceNet, ArcFace, and InsightFace, alongside privacy threat-modeling frameworks like **LINDDUN**, GDPR compliance requirements, serverless computing patterns, and state-of-the-art anti-spoofing techniques. [12]
- **Technology Stack Evaluation** covers a comparison of facial recognition libraries, embedding search solutions, cloud platforms, and message queue systems
- **Requirements Documentation** phase involves defining functional, non-functional, and security requirements and preparing the overall Project Specification Document.

8.1.2 System Architecture Design

- ***Microservices Architecture Design*** process outlines the overall structure of the Authentication Service, Biometric Processing Service, Identity Verification Service, and Data Management Service.
- ***Hexagonal Architecture*** component describes how ports and adapters are defined for external dependencies, together with the establishment of dependency injection patterns and inter-service communication protocols.
- ***Database & Security Design*** activities include the creation of a relational schema for user management, the organization of embedding storage in FAISS, Redis caching structures, and data-retention policies, as well as the planning of end-to-end encryption, API authentication mechanisms, and audit logging.

8.1.3 Core Services Development

- ***Authentication Service*** covers user registration with BCrypt hashing and the use of JWT-based authentication supported by refresh tokens.
- ***Biometric Processing Service*** integrates DeepFace, MediaPipe Face Mesh, an anti-spoofing module, and the active liveness detection pipeline.
- ***Identity Verification Service*** includes 1:1 verification (cosine similarity >0.6), 1:N identification (FAISS top-k), enrollment with ID parsing, and the maintenance of verification history.
- ***Data Management Service*** incorporates PostgreSQL repository pattern, FAISS index management, Redis caching, Kafka event publishing, and transactional consistency.

8.1.4 Frontend Development

- **Web Application** encompasses Authentication UI, enrollment wizard with ID upload and live capture, verification interface with real-time camera, admin dashboard, responsive design.
- **Mobile Application** includes native camera integration, real-time face detection overlay, enrollment flow, verification with challenge-response UI, push notifications, and offline capability.
- **Desktop Application** provides kiosk mode for self-service, admin interface for bulk operations, reporting with JFreeChart, and configuration management UI.

8.1.5 System Integration

- **API Gateway Configuration** involves the application of rate limiting, request routing, API key–based authentication, and CORS policies.
- **Message Queue Integration** includes the configuration of Kafka topics, the operation of producers and consumers, the use of RabbitMQ for task-queue workflows, and dead-letter queue handling.
- **Caching Strategy** includes Redis cluster configuration, the use of the cache-aside pattern, and the optimization of TTL parameters.

8.1.6 Testing & Quality Assurance

Functional Testing focuses on verifying the accuracy of face recognition, liveness detection, and identity-verification components across all modules to ensure correct behavior.

- **Performance Testing** evaluates system responsiveness, scalability, and stability under high-load scenarios to confirm operational efficiency.
- **Security Testing** incorporates penetration assessments and vulnerability analysis to ensure the robustness of data protection, encryption mechanisms, and system resilience.
- **Usability Testing** analyzes accessibility, interface clarity, and overall user experience as part of the user-acceptance evaluation process.

8.1.7 Deployment & Documentation

- **Production Deployment** describes the deployment of the system in a Kubernetes-based cloud environment featuring secure networking, database replication, and scalable service operation.
- ***Monitoring & Logging*** activities involve setting up monitoring dashboards and centralized logging systems to track performance, detect errors, and maintain system reliability.
- ***Backup & Disaster Recovery*** planning includes automated backup processes and resilience procedures designed to ensure data integrity and minimize downtime.

8.1.8 Final Testing & Presentation

- ***End-to-End Validation*** covers the testing of complete user journeys, the evaluation of failure scenarios, the verification of data consistency, and the confirmation that all acceptance criteria have been met.
- ***Deliverables Preparation*** includes finalizing source code, organizing deployment artifacts, completing all documentation, and preparing materials for the final presentation.

8.2. Division of Responsibilities

The project adopts a cross-functional team structure where all members contribute to every component while maintaining weighted responsibilities based on expertise. Ahmet Abdullah Gültekin serves as primary owner for backend development, Ayşe Gülsüm Eren drives application development, and Ayşenur Arıcı leads AI/ML implementation. However, each member actively participates in all project areas ensuring no single point of failure. This shared ownership model promotes knowledge distribution, enables parallel development, and maintains project continuity. Figure 5

illustrates the proportional contribution of each team member across the major project phases, reinforcing the balanced and collaborative nature of the team structure.

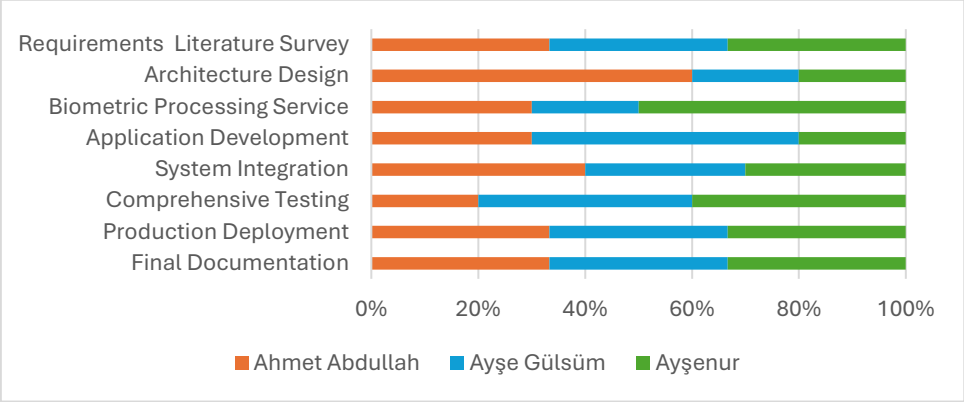

Figure 5:Task Distribution Chart

8.3. Timeline

As illustrated in *Figure 6*, the project timeline spans September to May, with key milestones marking progress across the eight phases. This schedule provides a clear roadmap for completing all development and integration activities on time.

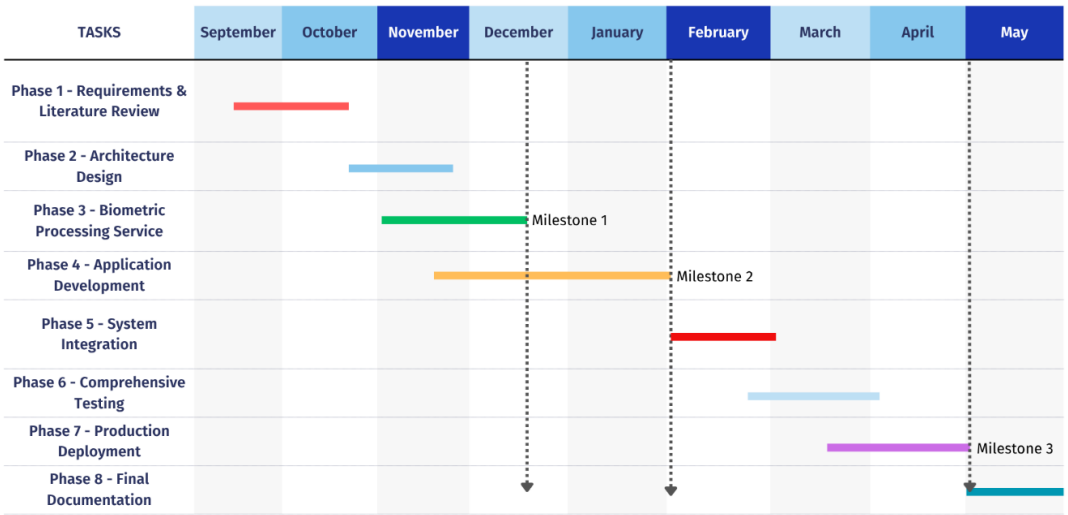

Figure 6: Project Timeline

9. SUCCESS FACTORS AND RISK MANAGEMENT

9.1. Measurability/Measuring Success

The success of the project will be evaluated based on the extent to which each objective defined in Section 3 is met. The success criteria for each objective are defined below:

- ***Objective 1:*** To develop an innovative and active liveness detection algorithm ("Biometric Puzzle").

  Success Factor: The developed algorithm can be completed by legitimate users with a success rate of over 95%, and it rejects simple spoofing attacks, such as static photos or videos held up to the camera, with a rate of over 99%.
- ***Objective 2:*** To establish a scalable, multi-tenant, cloud-native SaaS platform.

  Success Factor: All microservices run smoothly in a Dockerized environment, and all API endpoints must successfully pass Postman tests.
- ***Objective 3:*** To develop a user-friendly mobile application.

  Success Factor: The mobile application runs stably on both Android and iOS devices, and an end-user can complete the enrollment and authentication flow smoothly in 1 minute.
- ***Objective 4:*** To verify identities using various document types.

  Success Factor: The system must securely and efficiently match liveness-verified selfies with document-extracted facial embeddings while maintaining fast verification times.

9.2. Risk Management

Potential risks that may be encountered during the project and the contingency plans (Plan B) for these risks are listed below:

- ***Risk 1: Suboptimal Performance of Selected Recognition Model:*** The initially selected base model (VGG-Face) within the DeepFace framework may fail to meet the target accuracy thresholds or latency requirements under varying lighting conditions and angles.

  Plan B: Leverage the modular design of the system to switch to alternative State-of-the-Art (SOTA) models such as ArcFace or Google FaceNet. Since the biometric processing service is decoupled from the core logic (Hexagonal Architecture), integrating a more robust model will require minimal code refactoring.
- ***Risk 2:*** **Library Performance Issues:** Google MediaPipe fails to deliver adequate performance for real-time analysis on the low-end devices used for testing.

Plan B: Reduce the number of landmarks to be analyzed, decrease the resolution of the video stream, or reduce the number of steps in the "Biometric Puzzle" to lighten the processing load.

- ***Risk 3:*** **Integration Problems*:*** Unexpected issues arise in communication between microservices written in different languages (Java and Python) or with the mobile application.
  Plan B: Define API contracts (OpenAPI) in detail from the very beginning. Consider a stricter protocol for inter-service communication, such as gRPC.

10. BENEFITS AND IMPACT OF THE PROJECT

10.1. Benefits/Implications

The project will benefit both end-users and service providers. End-users will experience secure and seamless authentication without dealing with passwords or access cards. Service providers will lower operational costs, enhance security, and gain a competitive edge by offering a modern authentication solution.

10.2. Scientific Impact

The “Biometric Puzzle” algorithm introduces an innovative approach to active liveness detection. Analysis of its performance, accuracy, and spoofing resistance could make valuable contributions to the field.

10.3. Economic/Commercial/Social Impact

Designed not merely as an academic prototype but as a **market-ready commercial product**, this project delivers a robust **Cloud-based SaaS** platform. Its **multi-tenant architecture** is engineered to support high scalability and a subscription-based revenue model, establishing a solid foundation for a viable startup ecosystem. By bridging the gap between advanced biometrics and enterprise needs, the platform drives digital transformation, ensures high-assurance authentication, and proactively mitigates identity fraud.

10.4. Potential Impact on New Projects

Combining modern practices such as microservices, cloud-native technologies, and mobile machine learning, the project’s open-source infrastructure can guide future student projects. It may inspire new initiatives integrating other biometric modalities or developing adaptive authentication systems.

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